Shivam Mahajan

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SUMMARY

Software Engineer with 3+ years of experience building full-stack web applications, optimizing cloud infrastructure with AWS, and developing AI-powered applications and agent-based workflows. Proficient in Python, TypeScript, JavaScript, C#, React, AWS, and SQL, with strong expertise in algorithms, data structures, problem-solving, and building scalable, production-ready systems.

WORK EXPERIENCE

Software Engineer, Mindex | New York ***Oct 2023 – July 2026***

- Built AI agents using Github Copilot and MCP to automate code reviews, work-item creation, and sprint retro analysis, improving edge-case detection and cutting manual engineering and reporting effort by 40%
- Developed an integration platform using Angular, TypeScript, C# and AWS Lambda, orchestrated via CloudFormation and EventBridge to extract, transform, and synchronize data across third-party platforms
- Developed AI agents to automate engineering preplanning and implementation, generating technical plans in Azure DevOps, applying code changes, and reducing manual analysis and development effort by 60%
- Developed a Dashboard Web App using React, TypeScript, Python, AWS API Gateway, Lambda, and RDS, delivering insights to ~10 internal stakeholders and reducing analysis time by 50+ hours quarterly
- Translated ambiguous customer requirements into a technical solution by tracing URL update references across the codebase and implementing SQL automation and configuration changes, reducing manual effort by 50%
- Optimized JavaScript translation logic by batching API requests, improving performance by 80%

Software Engineer Intern, Mindex | New York ***Jan 2022 – Aug 2022***

- Spearheaded architectural improvements for a C#/.NET application hosted on AWS, using Redis caching and API design enhancements to decrease page load time by 30%
- Pioneered unit testing with NUnit and Moq, establishing the initial testing infrastructure that uncovered and resolved a critical bug impacting 16,000 users
- Architected an infrastructure with AWS CloudFormation and implemented serverless architectures using C#, Lambda, DynamoDB, decoupling microservices with SNS and SQS, reducing critical errors by 20%
- Collaborated with 11 developers using Agile principles, conducting code reviews that uncovered and rectified a major bug preventing updates to 1M+ database records

PROJECTS

Multi-Agent AI Research System | *Python, LangChain, OpenAI API*

- Developed a LangChain-based multi-agent research system that coordinates search and reader agents with writer and critic chains to collect web information and generate structured research reports
- Architected a critic-guided revision loop that improved report quality by 25%

AI-Powered Personal Tutor – Full Stack | *Node.js, Express, React, Tailwind CSS, OpenAI API and MongoDB*

- Developed an AI-driven web app using React, Redux, and Node.js to generate speech-enabled PDF summaries, boosting user engagement and learning outcomes
- Automated quiz generation from text summaries using the GPT-4 API, enhancing user retention by 30%

Discord Clone – Full Stack | *Next.js, React, TypeScript,* Tailwind CSS, *Prisma, PostgreSQL and Socket.io*

- Designed a real-time communication platform like Discord, using React custom hooks, Prisma with PostgreSQL for type-safe API queries, and Socket.io for messaging, voice, and video calls
- Deployed the app with Docker, AWS ECS, and ECR, automating delivery with a Jenkins CI/CD pipeline, accelerating update rollouts by 25%

EDUCATION

Master of Science in Computer Science ***GPA: 3.79/4.0***

Rochester Institute of Technology ***Aug 2020 – Aug 2023***

Relevant Coursework: Object-Oriented Programming, Data Structures & Algorithms, Web Development, Cloud Computing

TECHNICAL SKILLS

Programming Languages: Python, TypeScript, Java, JavaScript, PHP, C#, SQL

Web Technologies: React.js, Next.js, Express, AngularJS, Node.js, Flask, Django, AWS, HTML5, CSS, Docker, Git

Certifications: AWS Certified Developer – Associate, AWS Certified Solutions Architect – Associate